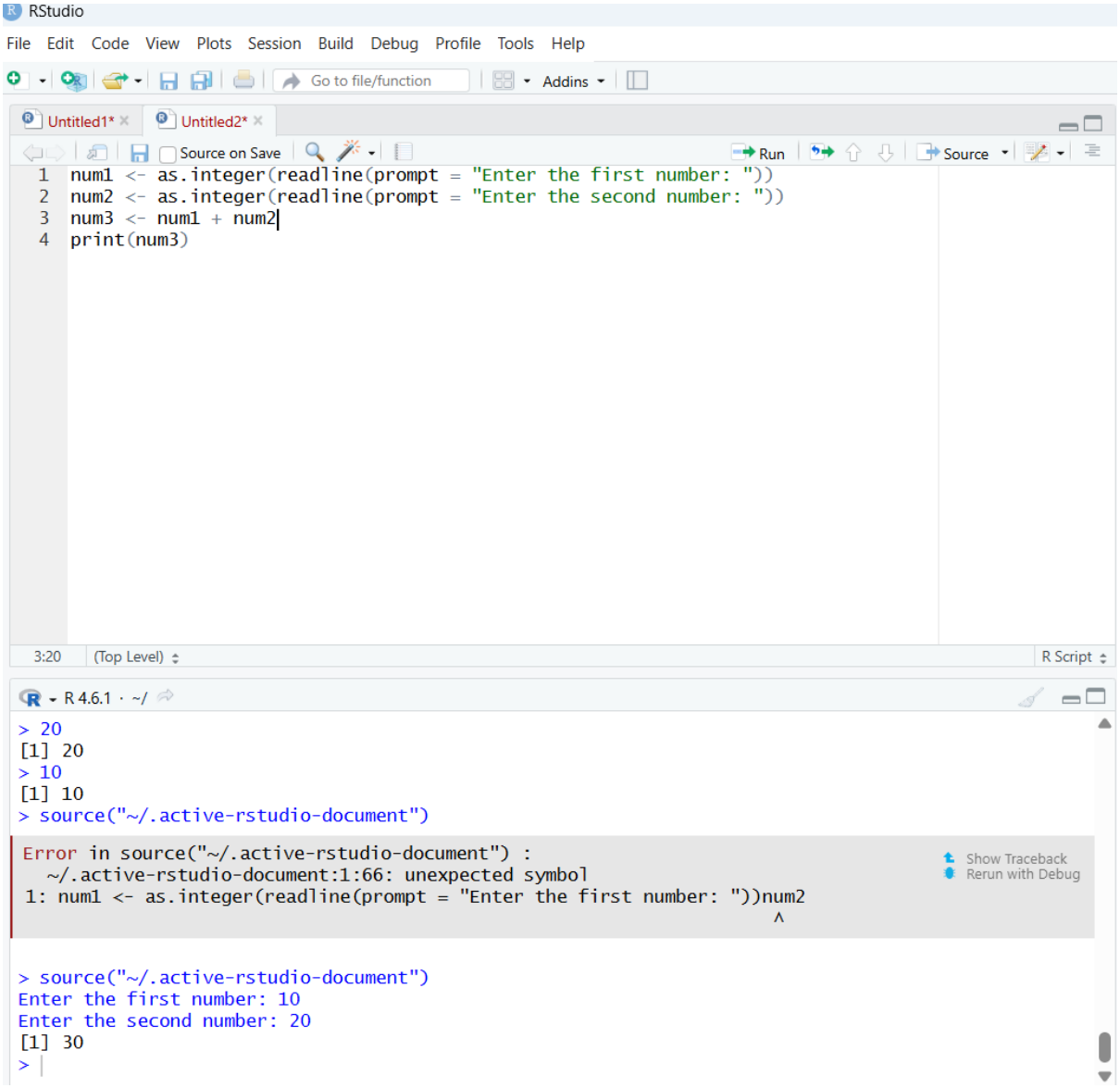
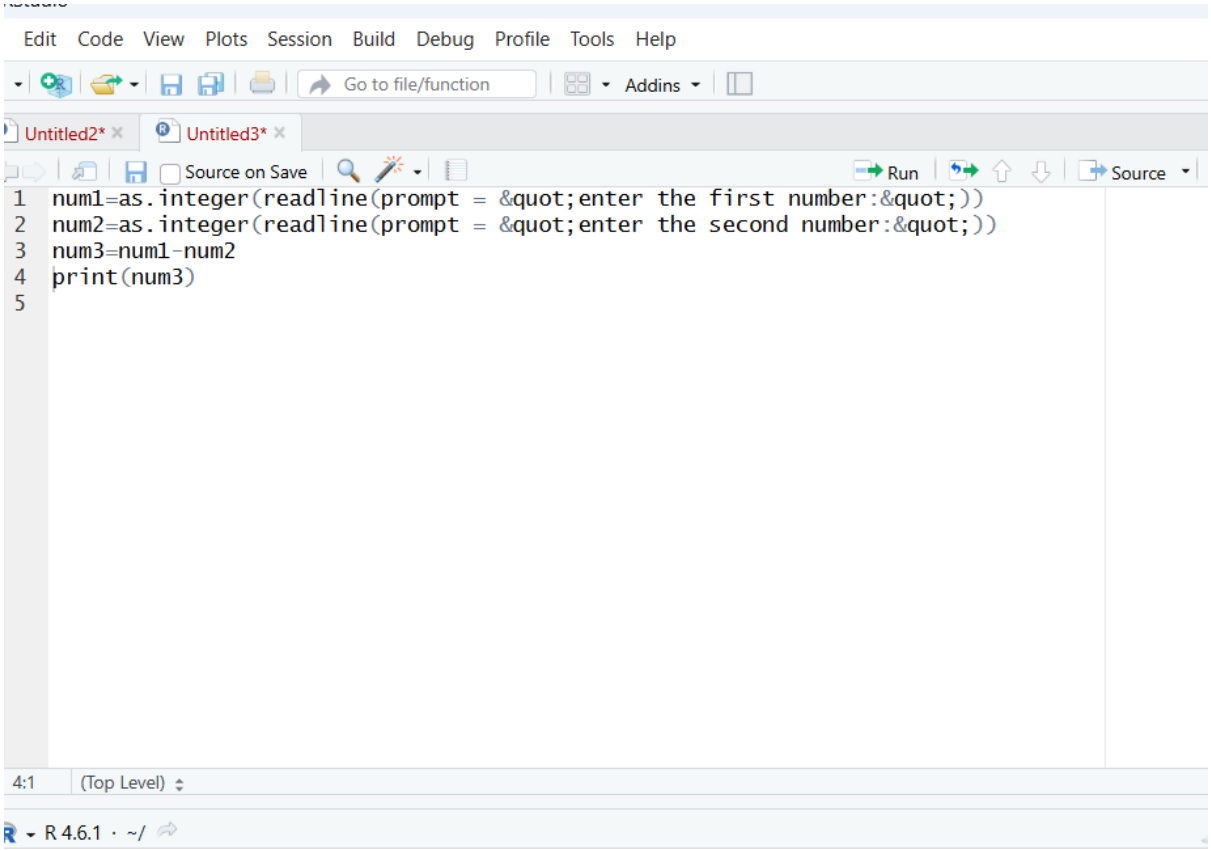
NAME:K.SUNIL KUMAR(192325091)

1)



2)



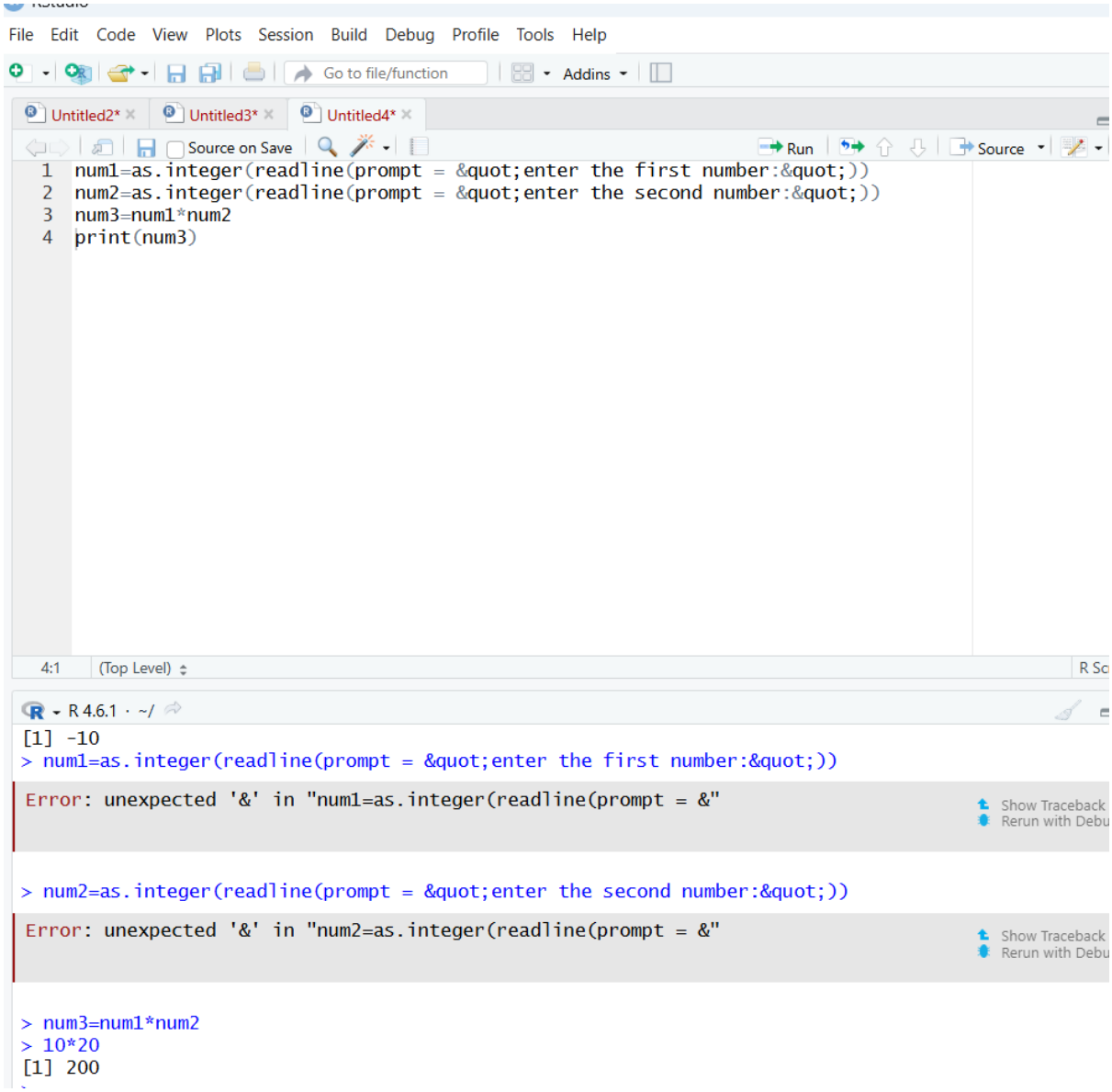

```
num1=as.integer(readline(prompt = &quot;enter the first number:&quot;))
rror: unexpected '&' in "num1=as.integer(readline(prompt = &"
```

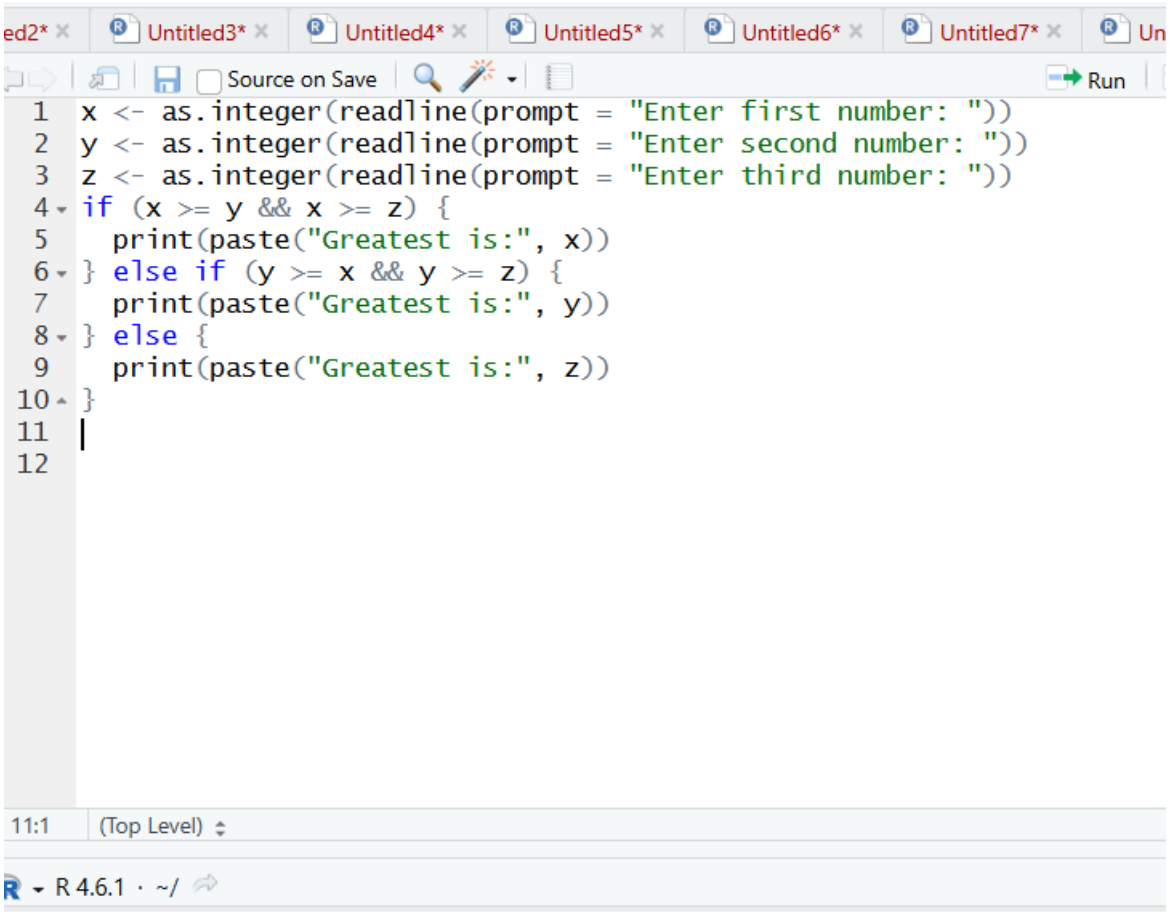
Show Tra
Rerun wit

```
num2=as.integer(readline(prompt = &quot;enter the second number:&quot;))
rror: unexpected '&' in "num2=as.integer(readline(prompt = &"
```

Show Tra
Rerun wit

```
num3=num1-num2
10-20
1] -10
```


File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled2*
Untitled3*
Untitled4*
Source on Save
Run
Source
1 num1=as.integer(readline(prompt = "enter the first number:"))
2 num2=as.integer(readline(prompt = "enter the second number:"))
3 num3=num1*num2
4 print(num3)
4:1
(Top Level)
R 4.6.1 · ~/
[1] -10
> num1=as.integer(readline(prompt = "enter the first number:"))
Error: unexpected '&' in "num1=as.integer(readline(prompt = &"
Show Traceback
Rerun with Debu
> num2=as.integer(readline(prompt = "enter the second number:"))
Error: unexpected '&' in "num2=as.integer(readline(prompt = &"
Show Traceback
Rerun with Debu
> num3=num1*num2
> 10*20
[1] 200



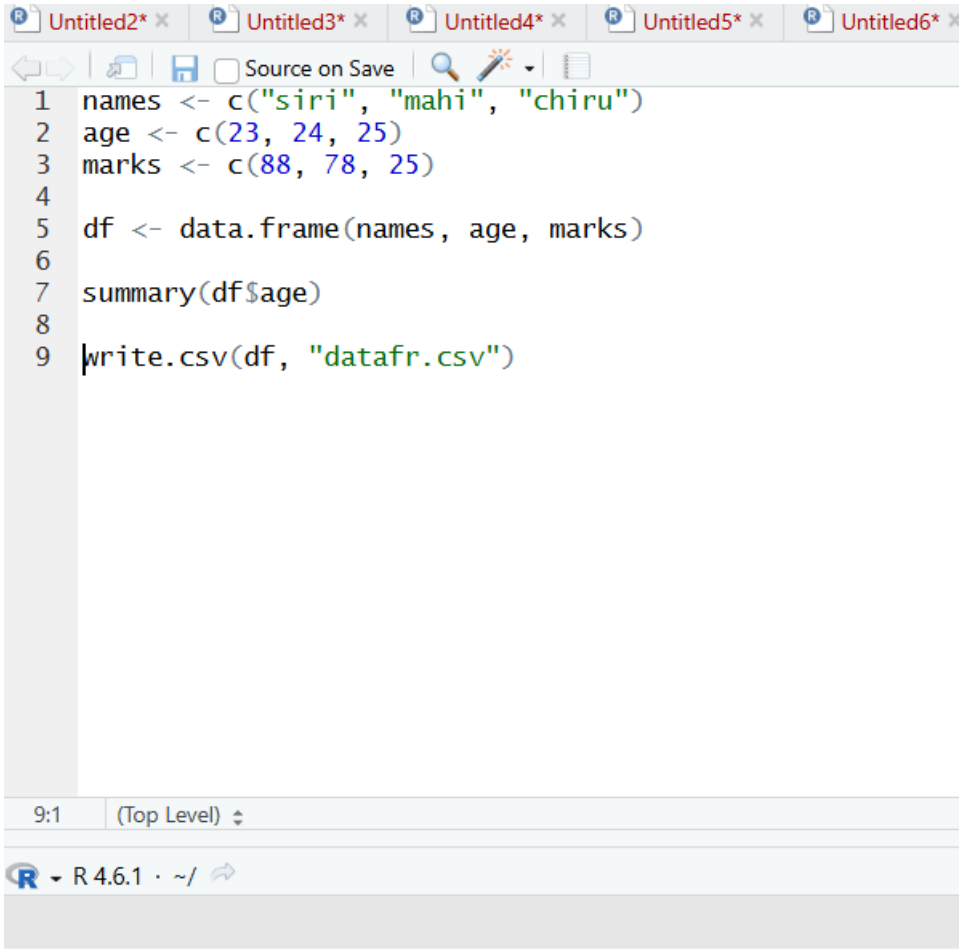

```
 z <- as.integer(readline(prompt = "Enter third number: "))
nter third number: if (x >= y && x >= z) {
```

```
Warning message:
NAs introduced by coercion
```

```
   print(paste("Greatest is:", x))
1] "Greatest is: 10"
 } else if (y >= x && y >= z) {
```

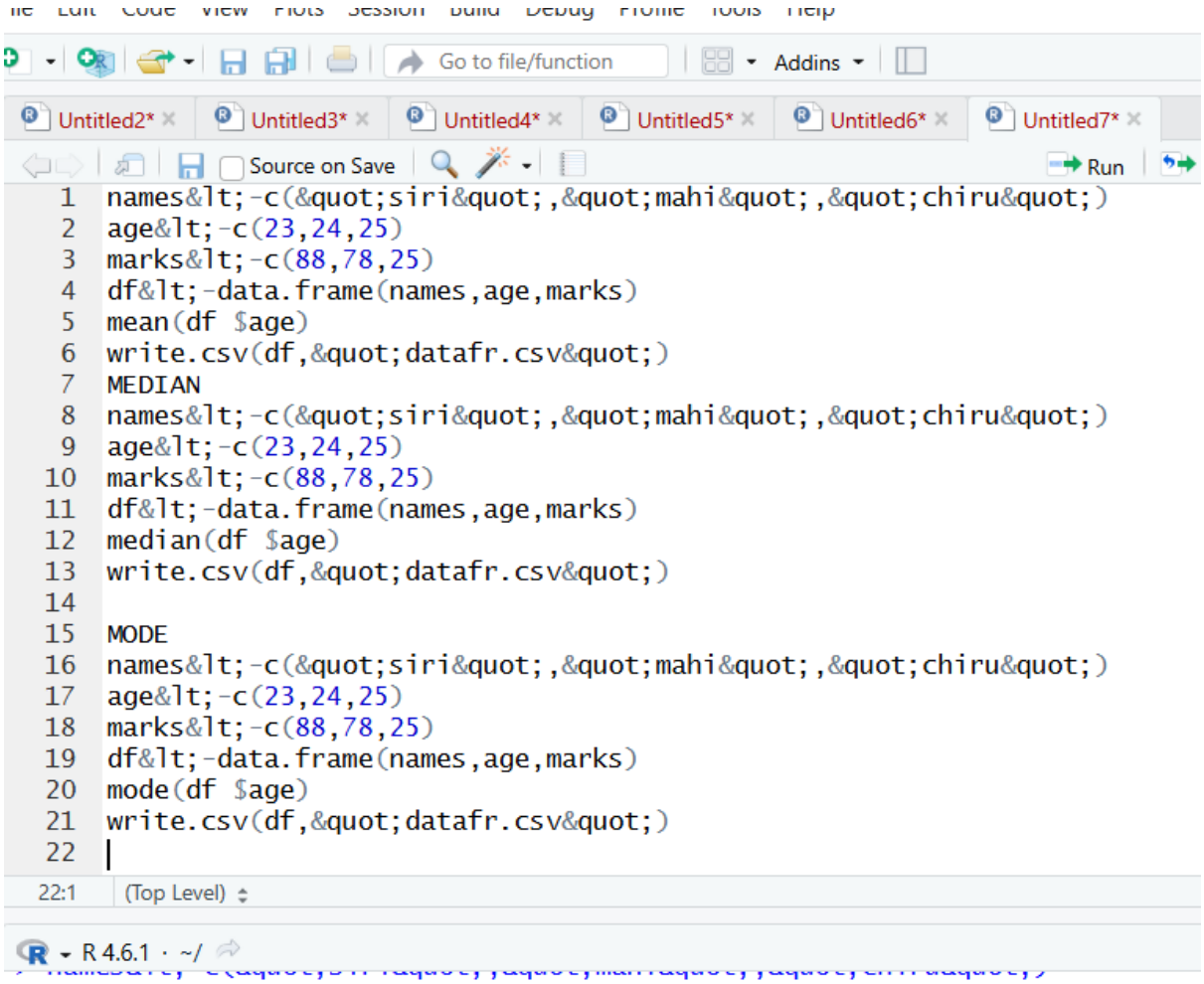
```
Error: unexpected '}' in "}"
```

```
 30
```




```
> write.csv(df,&amp;quot;datafr.csv&amp;quot;)
Error: unexpected '&' in "write.csv(df,&"

> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> summary(df$age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   23.0    23.5    24.0    24.0    24.5    25.0
>
```

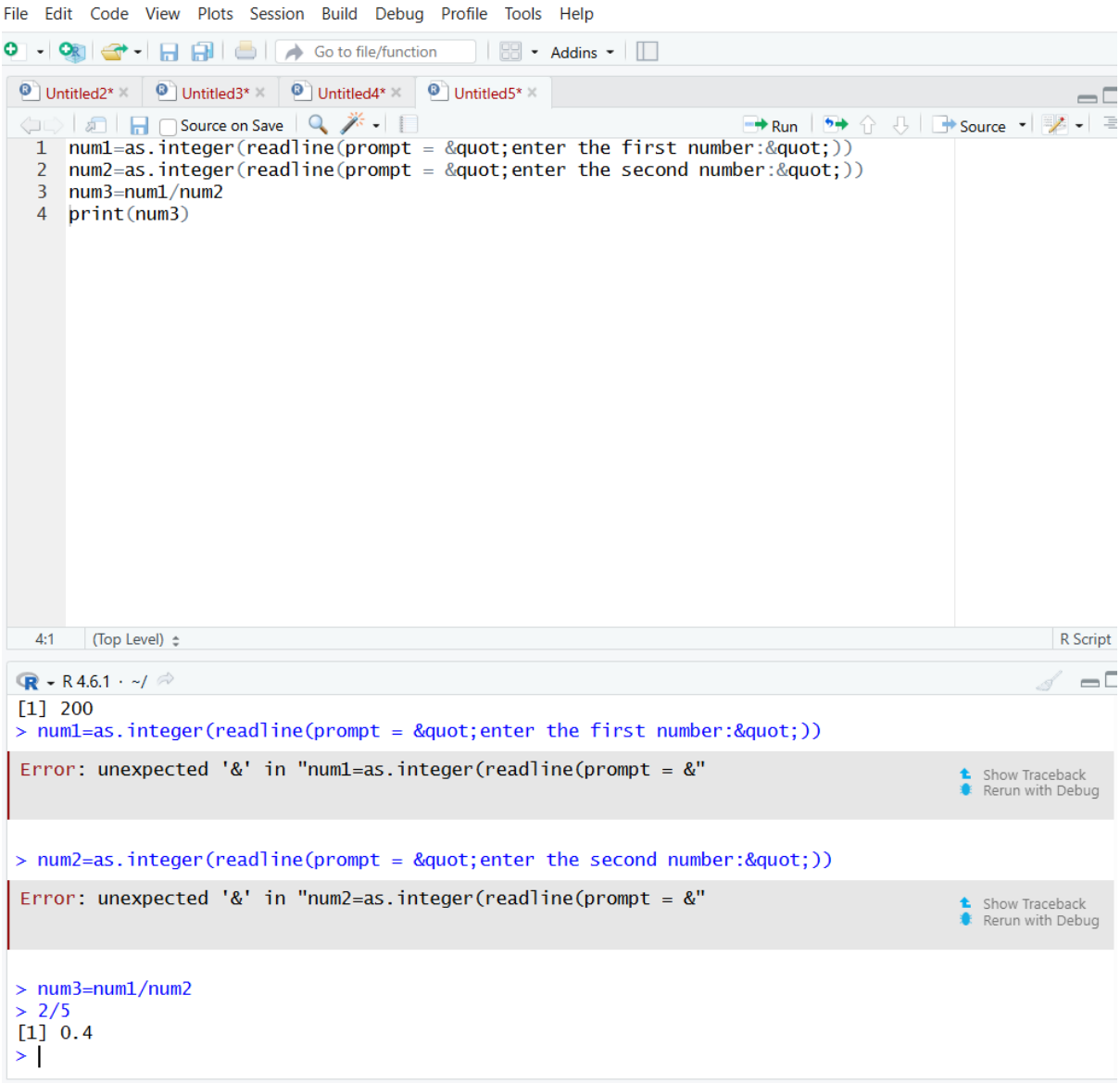


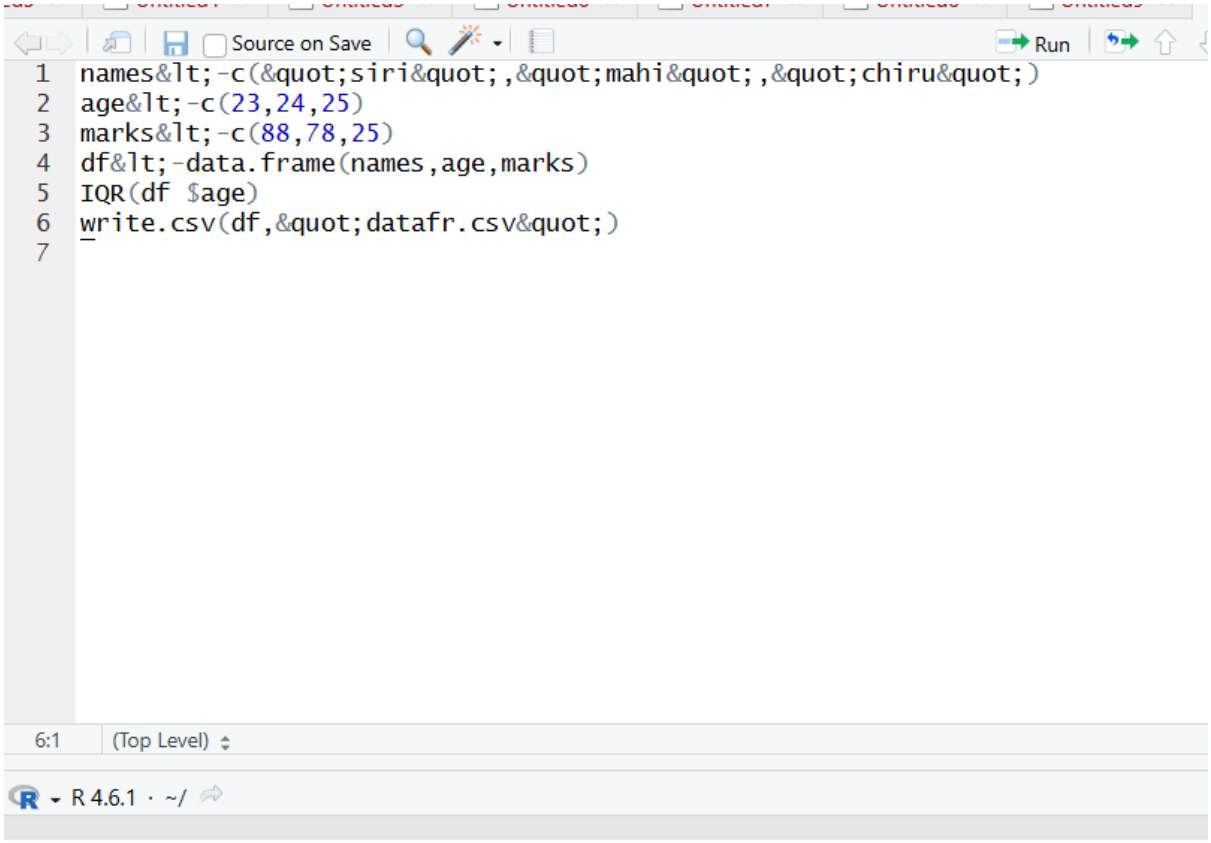

```
Error: object 'lt' not found
```

```
> age&lt;-c(23,24,25)
Error: object 'lt' not found
```

```
> marks&lt;-c(88,78,25)
Error: object 'marks' not found
```


File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled2*
Untitled3*
Untitled4*
Untitled5*
Untitled6*
Source on Save
1 num <- as.integer(readline(prompt = "Enter a number: "))
2
3 if (num %% 2 == 0) {
4 print("Number is even")
5 } else {
6 print("Number is odd")
7 }
8
9
8:1
(Top Level)
R 4.6.1 · ~/
> print("Number is even")
[1] "Number is even"
> } else {
Error: unexpected '}' in "}"
>
>
>
>
> source("~/.active-rstudio-document")
Enter a number: 4
[1] "Number is even"
>


File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled2*
Untitled3*
Untitled4*
Untitled5*
Source on Save
Run
Source
1 num1=as.integer(readline(prompt = "enter the first number:"))
2 num2=as.integer(readline(prompt = "enter the second number:"))
3 num3=num1/num2
4 print(num3)
4:1
(Top Level)
R Script
R 4.6.1 · ~/
[1] 200
> num1=as.integer(readline(prompt = "enter the first number:"))
Error: unexpected '&' in "num1=as.integer(readline(prompt = &"
Show Traceback
Rerun with Debug
> num2=as.integer(readline(prompt = "enter the second number:"))
Error: unexpected '&' in "num2=as.integer(readline(prompt = &"
Show Traceback
Rerun with Debug
> num3=num1/num2
> 2/5
[1] 0.4
>




```
> marks&lt;-c(88,78,25)
Error: object 'lt' not found

> df&lt;-data.frame(names,age,marks)
Error: object 'lt' not found

> IQR(df $age)
[1] 1
>
```

```r
# Linear Regression
Relation <- lm(diabetes$BloodPressure ~ diabetes$Age)

# Save plot as PNG file
png(filename = "linear_regression.png")

plot(diabetes$Age, diabetes$BloodPressure,
     col = "green",
     main = "Linear Regression Analysis",
     xlab = "Age",
     ylab = "Blood Pressure")

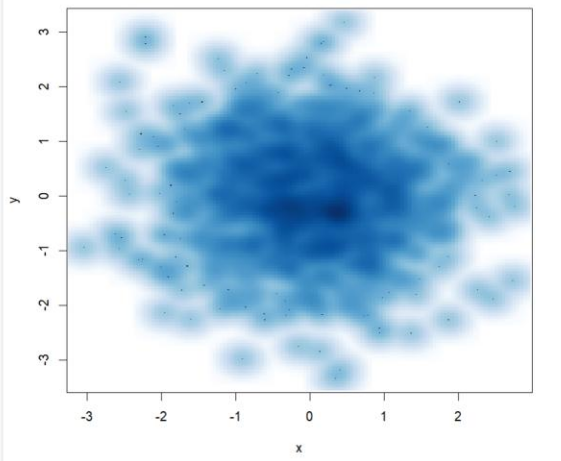
abline(Relation, col = "red")

dev.off()

# Display regression result
print(summary(Relation))
```

```
R 4.6.1 · ~/
# Linear Regression
Relation <- lm(diabetes$BloodPressure ~ diabetes$Age)
Error in eval(predvars, data, env) : object 'diabetes' not found
Show Traceback
Rerun with Debug

# Save plot as PNG file
png(filename = "linear_regression.png")
plot(diabetes$Age, diabetes$BloodPressure,
     col = "green",
     main = "Linear Regression Analysis",
     xlab = "Age",
     ylab = "Blood Pressure")
Error: object 'diabetes' not found
Show Traceback
Rerun with Debug
```



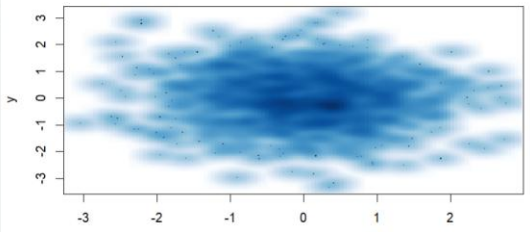

```r
set.seed(9)

x <- rnorm(1000)
y <- rnorm(1000)

smoothScatter(y ~ x)
smoothScatter(x, y)
```

```
R 4.6.1 · ~/
> set.seed(9)
> x <- rnorm(1000)
> y <- rnorm(1000)
> smoothScatter(y ~ x)
> smoothScatter(x, y)
>
```

Q3	Named num 35
sd_val	316.227766016838
smoothed_boundary	Named num [1:52] 1 1 5 5 5 5 5 8 8 10 ...
smoothed_mean	Named num [1:52] 2.33 2.33 2.33 5 5 ...
Std	9.86576572463249
upper	num [1:6] 5 15 20 50 80 110
v	80000
v_norm	0.6
x	num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ...
y	num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ...



Untitled10* | Untitled13* | Untitled14* | Untitled15* | Untitled16* | Untitled17*

Source on Save | Run

```
1
2  library(readxl)
3
4  diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
5
6  correlation <- cor(diabetest1$Age,
7                     diabetest1$Insulin,
8                     use = "complete.obs")
9
10 print(correlation)
11
12
```

2:1 (Top Level)

R 4.6.1 · ~/

```
> library(readxl)
Error in library(readxl) : there is no package called 'readxl'

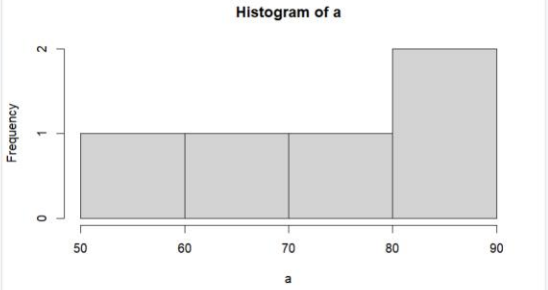
> diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
Error in read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx") :
  could not find function "read_excel"

> correlation <- cor(diabetest1$Age,
+                    diabetest1$Insulin,
+                    use = "complete.obs")
```

```
a <- c(55, 67, 89, 80, 90)
hist(a)
```

```
> a <- c(55, 67, 89, 80, 90)
> hist(a)
>
```

df	3 obs. of 3 variables
Values	
a	num [1:5] 55 67 89 80 90
A	num [1:3] 18 2 20
age	num [1:3] 23 24 25
B	num [1:3] 22 28 10
C	num [1:3] 20 40 40
c_prev	950
cf	num [1:6] 200 650 950 2450 3150 ...
classA	num [1:9] 76 35 47 64 95 66 89 36 84




```
names <- c("siri", "chru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)

df <- data.frame(names, age, marks)

# Histogram
hist(df$age, main = "Histogram of Age",
     xlab = "Age", col = "skyblue")

# Box Plot
boxplot(df$age, main = "Box Plot of Age",
        ylab = "Age", col = "lightgreen")

```

```
> names <- c("siri", "chru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> # Histogram
> hist(df$age, main = "Histogram of Age",
+      xlab = "Age", col = "skyblue")
> # Box Plot
> boxplot(df$age, main = "Box Plot of Age",
+         ylab = "Age", col = "lightgreen")
>
>
```

df	3 obs. of 3 variables
Values	
a	num [1:5] 55 67 89 80 90
A	num [1:3] 18 2 20
age	num [1:3] 23 24 25
B	num [1:3] 22 28 10
C	num [1:3] 20 40 40
c_prev	950
cf	num [1:6] 200 650 950 2450 3150 ...
classA	num [1:9] 76 35 47 64 95 66 89 36 84



```r
names <- c("siri", "chru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)

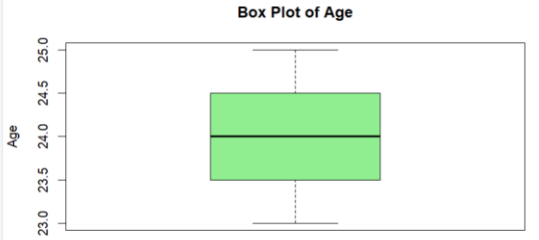
df <- data.frame(names, age, marks)

# Histogram
hist(df$age, main = "Histogram of Age",
     xlab = "Age", col = "skyblue")

# Box Plot
boxplot(df$age, main = "Box Plot of Age",
        ylab = "Age", col = "lightgreen")
```

```
> names <- c("siri", "chru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> # Histogram
> hist(df$age, main = "Histogram of Age",
+      xlab = "Age", col = "skyblue")
> # Box Plot
> boxplot(df$age, main = "Box Plot of Age",
+         ylab = "Age", col = "lightgreen")
>
>
```

Values	
df	3 obs. of 3 variables
a	num [1:5] 55 67 89 80 90
A	num [1:3] 18 2 20
age	num [1:3] 23 24 25
B	num [1:3] 22 28 10
C	num [1:3] 20 40 40
c_prev	950
cf	num [1:6] 200 650 950 2450 3150 ...
classA	num [1:9] 76 35 47 64 95 66 89 36 84

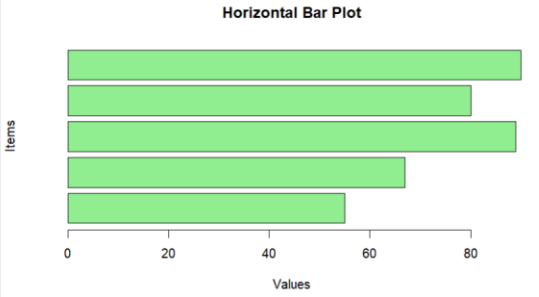



```r
# Vertical Bar Plot
a <- c(55, 67, 89, 80, 90)
barplot(a, main = "Vertical Bar Plot",
        xlab = "Items", ylab = "Values",
        col = "skyblue")

# Horizontal Bar Plot
a <- c(55, 67, 89, 80, 90)
barplot(a, horiz = TRUE,
        main = "Horizontal Bar Plot",
        xlab = "Values", ylab = "Items",
        col = "lightgreen")
```

```
> # Vertical Bar Plot
> a <- c(55, 67, 89, 80, 90)
> barplot(a, main = "Vertical Bar Plot",
+         xlab = "Items", ylab = "Values",
+         col = "skyblue")
> # Horizontal Bar Plot
> a <- c(55, 67, 89, 80, 90)
> barplot(a, horiz = TRUE,
+         main = "Horizontal Bar Plot",
+         xlab = "Values", ylab = "Items",
+         col = "lightgreen")
>
```

Values	
a	num [1:5] 55 67 89 80 90
A	num [1:3] 18 2 20
age	num [1:3] 23 24 25
B	num [1:3] 22 28 10
C	num [1:3] 20 40 40
c_prev	950
cf	num [1:6] 200 650 950 2450 3150 ...
classA	num [1:9] 76 35 47 64 95 66 89 36 84
classB	num [1:9] 51 56 84 60 59 70 63 66 50



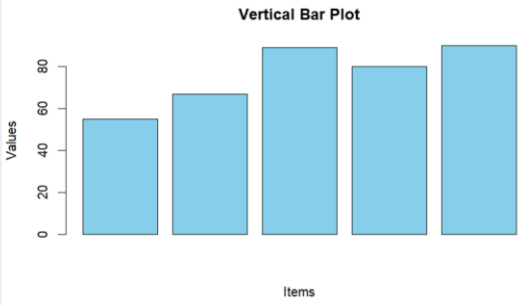
```r
# Vertical Bar Plot
a <- c(55, 67, 89, 80, 90)
barplot(a, main = "Vertical Bar Plot",
        xlab = "Items", ylab = "Values",
        col = "skyblue")

# Horizontal Bar Plot
a <- c(55, 67, 89, 80, 90)
barplot(a, horiz = TRUE,
        main = "Horizontal Bar Plot",
        xlab = "Values", ylab = "Items",
        col = "lightgreen")
```

```
> # Vertical Bar Plot
> a <- c(55, 67, 89, 80, 90)
> barplot(a, main = "Vertical Bar Plot",
+         xlab = "Items", ylab = "Values",
+         col = "skyblue")
> # Horizontal Bar Plot
> a <- c(55, 67, 89, 80, 90)
> barplot(a, horiz = TRUE,
+         main = "Horizontal Bar Plot",
+         xlab = "Values", ylab = "Items",
+         col = "lightgreen")
>
```

Values

a	num [1:5] 55 67 89 80 90
A	num [1:3] 18 2 20
age	num [1:3] 23 24 25
B	num [1:3] 22 28 10
C	num [1:3] 20 40 40
c_prev	950
cf	num [1:6] 200 650 950 2450 3150 ...
classA	num [1:9] 76 35 47 64 95 66 89 36 84
classB	num [1:9] 51 56 84 60 59 70 63 66 50



File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

Untitled8* | Untitled9* | Untitled11* | Untitled12* | Untitled10* | Untitled13*

Source on Save | Run

```
library(readxl)

diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")

A <- c(diabetest1$Age)

# Mean
Mean <- mean(A, na.rm = TRUE)
print(Mean)

# Minimum
Minimum <- min(A, na.rm = TRUE)
print(Minimum)

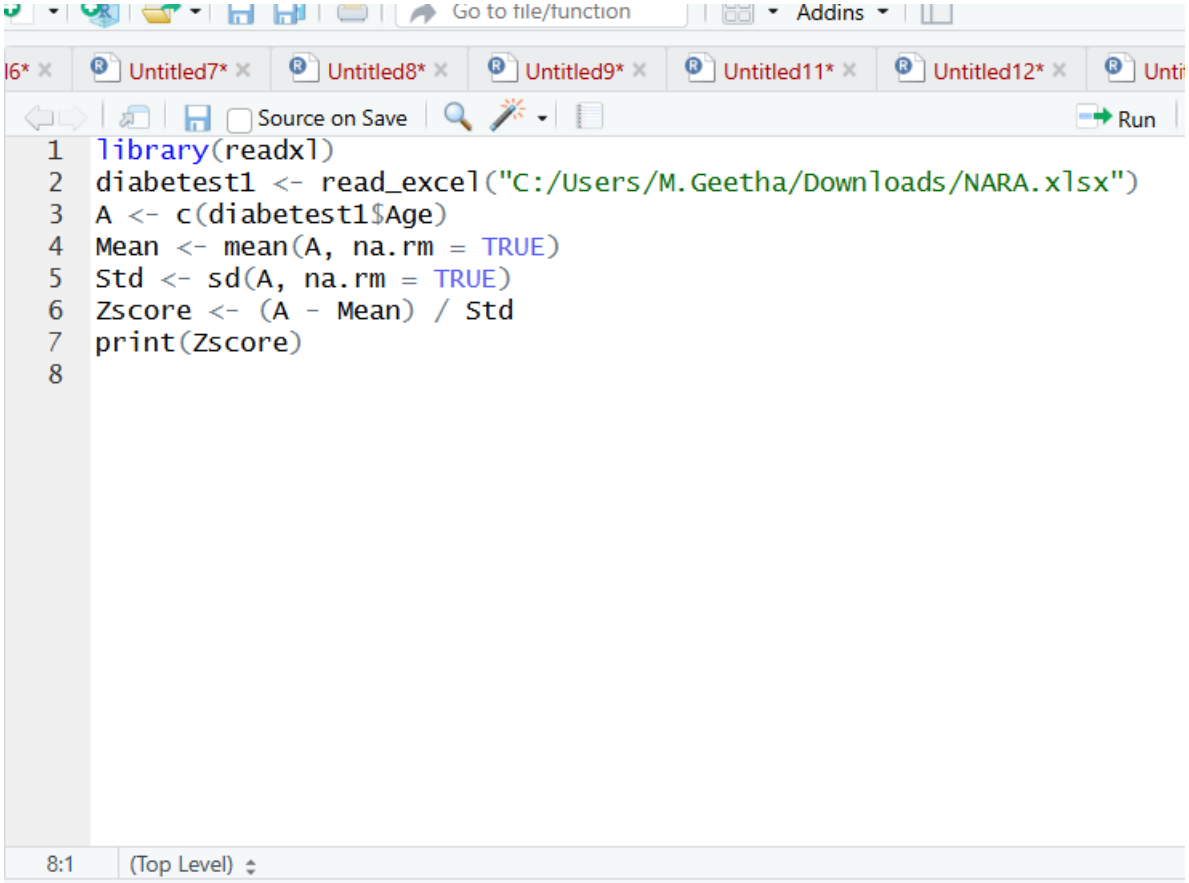
# Maximum
Maximum <- max(A, na.rm = TRUE)
print(Maximum)

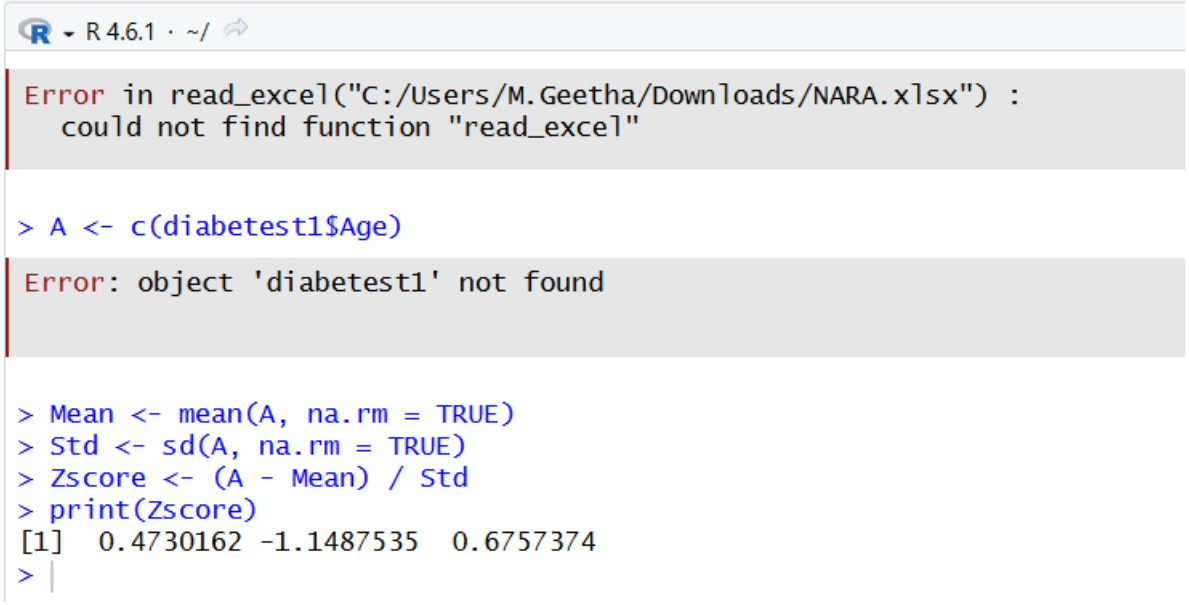
# Min-Max Normalization
MinMax <- (A - Minimum) / (Maximum - Minimum)
print(MinMax)
```

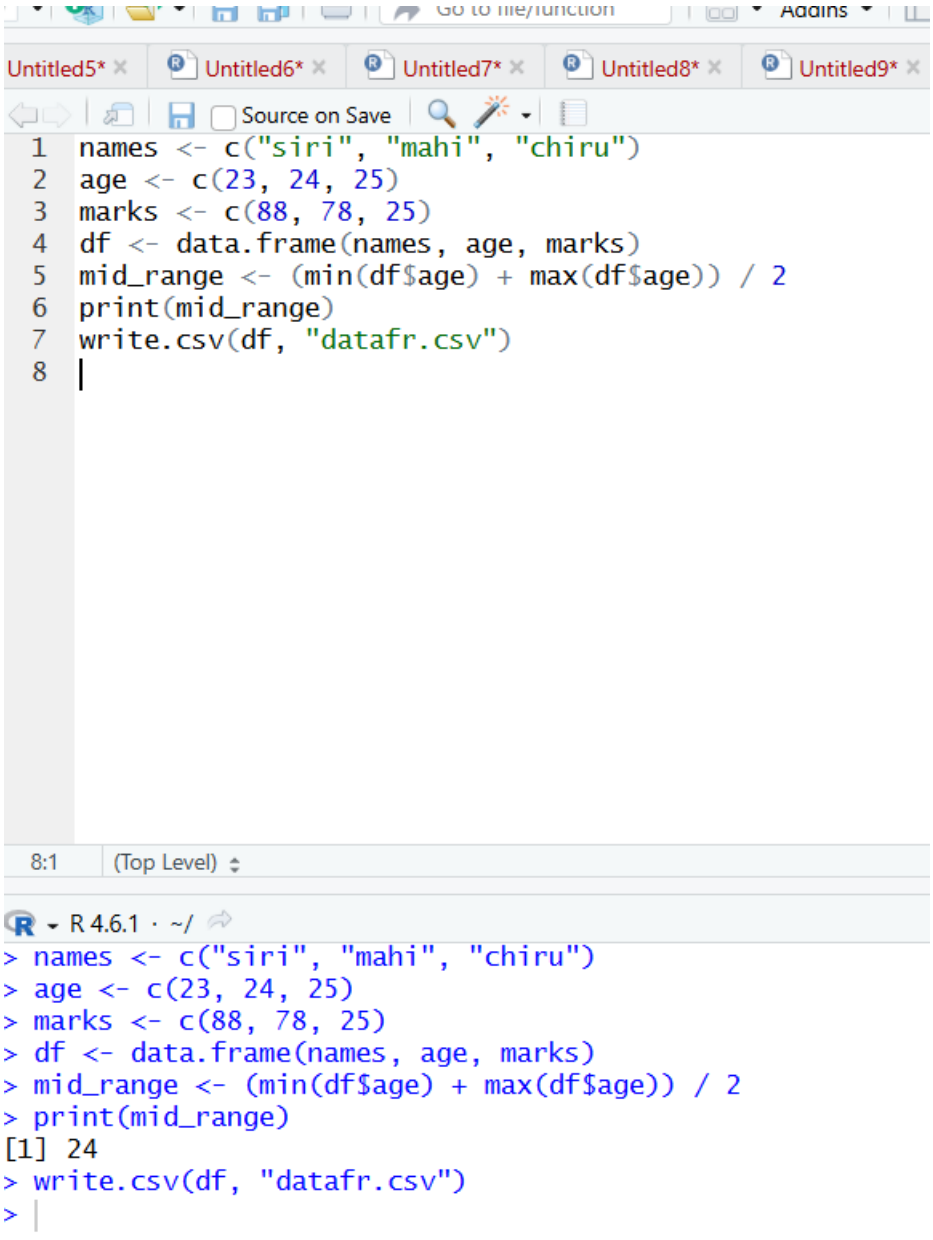
23:1 (Top Level)

R 4.6.1 · ~/

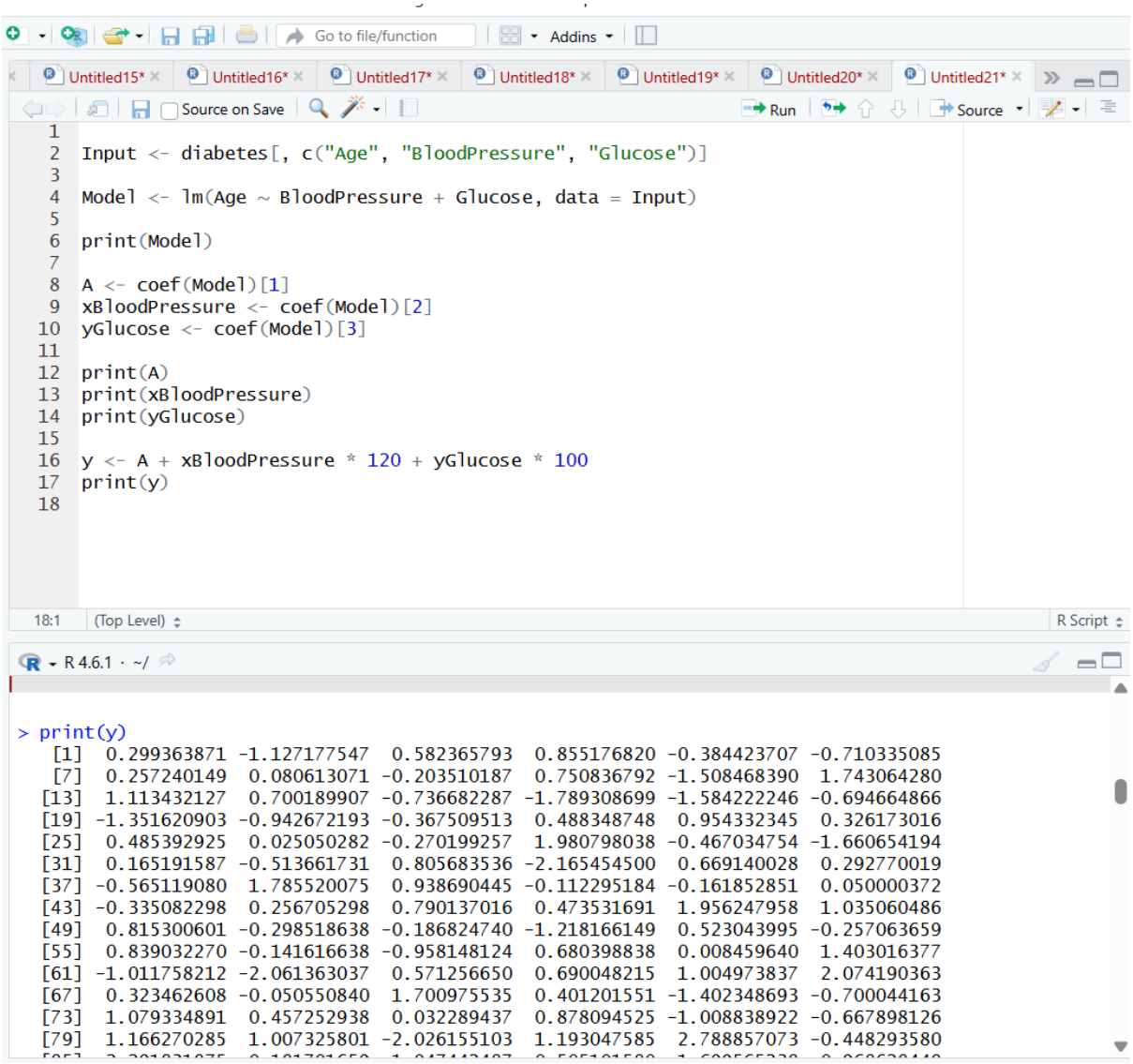
```
> # Mean
> Mean <- mean(A, na.rm = TRUE)
> print(Mean)
[1] 13.33333
> # Minimum
> Minimum <- min(A, na.rm = TRUE)
> print(Minimum)
[1] 2
> # Maximum
> Maximum <- max(A, na.rm = TRUE)
> print(Maximum)
[1] 20
> # Min-Max Normalization
> MinMax <- (A - Minimum) / (Maximum - Minimum)
> print(MinMax)
[1] 0.8888889 0.0000000 1.0000000
```


Go to file/function
Addins
Untitled7*
Untitled8*
Untitled9*
Untitled11*
Untitled12*
Source on Save
Run
1 library(readxl)
2 diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
3 A <- c(diabetest1$Age)
4 Mean <- mean(A, na.rm = TRUE)
5 Std <- sd(A, na.rm = TRUE)
6 Zscore <- (A - Mean) / Std
7 print(Zscore)
8
8:1
(Top Level)


R 4.6.1 · ~/
Error in read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx") :
could not find function "read_excel"
> A <- c(diabetest1$Age)
Error: object 'diabetest1' not found
> Mean <- mean(A, na.rm = TRUE)
> Std <- sd(A, na.rm = TRUE)
> Zscore <- (A - Mean) / Std
> print(Zscore)
[1] 0.4730162 -1.1487535 0.6757374
>


Addins
Untitled5*
Untitled6*
Untitled7*
Untitled8*
Untitled9*
Source on Save
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4 df <- data.frame(names, age, marks)
5 mid_range <- (min(df$age) + max(df$age)) / 2
6 print(mid_range)
7 write.csv(df, "datafr.csv")
8
8:1
(Top Level)
R 4.6.1 · ~/
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> mid_range <- (min(df$age) + max(df$age)) / 2
> print(mid_range)
[1] 24
> write.csv(df, "datafr.csv")
>


Untitled4*
Untitled5*
Untitled6*
Untitled7*
Untitled8*
Untitled9*
Source on Save
Run
Go back to the previous source location (Ctrl+F9)
ot;mahi","chiru")
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
quantile(df $age)
write.csv(df,"datafr.csv")
6:1
(Top Level)
R 4.6.1 · ~/
> marks<-c(88,78,25)
Error: object 'lt' not found
> df<-data.frame(names,age,marks)
Error: object 'lt' not found
> quantile(df $age)
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
>


Go to file/function
Addins
Untitled15*
Untitled16*
Untitled17*
Untitled18*
Untitled19*
Untitled20*
Untitled21*
Source on Save
Run
Source
1 Input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
2
3 Model <- lm(Age ~ BloodPressure + Glucose, data = Input)
4
5 print(Model)
6
7 A <- coef(Model)[1]
8 xBloodPressure <- coef(Model)[2]
9 yGlucose <- coef(Model)[3]
10
11 print(A)
12 print(xBloodPressure)
13 print(yGlucose)
14
15 y <- A + xBloodPressure * 120 + yGlucose * 100
16 print(y)
17
18
18:1
(Top Level)
R Script
R 4.6.1 · ~/
> print(y)
[1] 0.299363871 -1.127177547 0.582365793 0.855176820 -0.384423707 -0.710335085
[7] 0.257240149 0.080613071 -0.203510187 0.750836792 -1.508468390 1.743064280
[13] 1.113432127 0.700189907 -0.736682287 -1.789308699 -1.584222246 -0.694664866
[19] -1.351620903 -0.942672193 -0.367509513 0.488348748 0.954332345 0.326173016
[25] 0.485392925 0.025050282 -0.270199257 1.980798038 -0.467034754 -1.660654194
[31] 0.165191587 -0.513661731 0.805683536 -2.165454500 0.669140028 0.292770019
[37] -0.565119080 1.785520075 0.938690445 -0.112295184 -0.161852851 0.050000372
[43] -0.335082298 0.256705298 0.790137016 0.473531691 1.956247958 1.035060486
[49] 0.815300601 -0.298518638 -0.186824740 -1.218166149 0.523043995 -0.257063659
[55] 0.839032270 -0.141616638 -0.958148124 0.680398838 0.008459640 1.403016377
[61] -1.011758212 -2.061363037 0.571256650 0.690048215 1.004973837 2.074190363
[67] 0.323462608 -0.050550840 1.700975535 0.401201551 -1.402348693 -0.700044163
[73] 1.079334891 0.457252938 0.032289437 0.878094525 -1.008838922 -0.667898126
[79] 1.166270285 1.007325801 -2.026155103 1.193047585 2.788857073 -0.448293580